

30.3 Group penalties

A factor with k levels enters a linear model as $k - 1$ indicator columns (Section 8.5, Definition 8.5.1), and a smooth effect as several basis columns. The question is whether *the factor* matters. The lasso on the indicator columns can keep some levels and drop others, and its answer depends on the coding. Yuan and Lin (2006) proposed a penalty that treats a block of columns as a unit.

30.3.1 The group lasso

Partition the columns of $\mathbf{X}$ into groups $g = 1, \dots, G$, with $\mathbf{X}_g$ the $n \times p_g$ block of group g and $\mathbf{b}_g$ the matching block of coefficients, so that $\mathbf{X}\mathbf{b} = \sum_g \mathbf{X}_g \mathbf{b}_g$.

Definition 30.3.1. Group lasso

For weights $w_g > 0$, the **group lasso** criterion is

$$Q(\mathbf{b}) = \frac{1}{2} \left\| \mathbf{y} - \sum_{g=1}^G \mathbf{X}_g \mathbf{b}_g \right\|^2 + \lambda \sum_{g=1}^G w_g \|\mathbf{b}_g\|, \quad (30.3.1)$$

with Euclidean norms $\|\mathbf{b}_g\|$, not squared. The usual weights are $w_g = \sqrt{p_g}$. The columns of each group are usually centred and **orthonormalized within the group** before fitting, so that $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}_{p_g}$. Different groups are not made orthogonal to each other.

With singleton groups and unit weights this is the lasso. In general it acts like the lasso *across* groups and like ridge *within* them: $\|\mathbf{b}_g\|$ has a corner only at $\mathbf{b}_g = \mathbf{0}$, so a group is either absent or present with, typically, all its coefficients nonzero.

Proposition 30.3.1. Group lasso and block soft thresholding

- (a) A minimizer of Eq. 30.3.1 exists, and all minimizers have the same fitted vector.
- (b) (KKT conditions.) $\hat{\boldsymbol{\beta}}$ is a minimizer iff, with $\mathbf{r} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$, for every group g ,

$$\mathbf{X}_g^\top \mathbf{r} = \lambda w_g \frac{\hat{\boldsymbol{\beta}}_g}{\|\hat{\boldsymbol{\beta}}_g\|} \quad \text{if } \hat{\boldsymbol{\beta}}_g \neq \mathbf{0}, \quad \|\mathbf{X}_g^\top \mathbf{r}\| \leq \lambda w_g \quad \text{if } \hat{\boldsymbol{\beta}}_g = \mathbf{0}. \quad (30.3.2)$$

- (c) Let $\lambda > 0$. Then $\hat{\boldsymbol{\beta}} = \mathbf{0}$ iff $\lambda \geq \lambda_{\max} = \max_g \|\mathbf{X}_g^\top \mathbf{y}\|/w_g$.
- (d) (Orthonormal groups.) If $\mathbf{X}_g^\top \mathbf{X}_h = \mathbf{I}_{p_g}$ for $g = h$ and $\mathbf{0}$ for $g \neq h$, the unique minimizer is **block soft thresholding** of $\mathbf{z}_g = \mathbf{X}_g^\top \mathbf{y}$:

$$\hat{\boldsymbol{\beta}}_g = \left(1 - \frac{\lambda w_g}{\|\mathbf{z}_g\|}\right)_+ \mathbf{z}_g. \quad (30.3.3)$$

- (e) (Block update; compare Proposition 30.2.2.) If $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}_{p_g}$, then minimizing Eq. 30.3.1 over $\mathbf{b}_g$ with the other blocks fixed gives Eq. 30.3.3 with $\mathbf{z}_g$ replaced by $\mathbf{s}_g = \mathbf{X}_g^\top \mathbf{r}_{(g)}$, where $\mathbf{r}_{(g)} = \mathbf{y} - \sum_{h \neq g} \mathbf{X}_h \mathbf{b}_h$ is the partial residual.

Proof.

- (a) The penalty is convex and continuous. Since $\sum_g \|\mathbf{b}_g\| \geq \|\mathbf{b}\|$, it is at least $(\min_g w_g) \|\mathbf{b}\|$. Apply Theorem 30.1.1(a)(ii) and (b).
- (b) Combine Lemma 30.1.2 with Lemma 30.1.3, using the groups as the blocks.
- (c) At $\hat{\boldsymbol{\beta}} = \mathbf{0}$, the residual is $\mathbf{y}$, and Eq. 30.3.2 says that $\|\mathbf{X}_g^\top \mathbf{y}\| \leq \lambda w_g$ for every g . If this holds, $\mathbf{0}$ is a minimizer. Every minimizer then has fitted vector $\mathbf{0}$ and, by Theorem 30.1.1(b), penalty 0, so it is $\mathbf{0}$. If it fails, $\mathbf{0}$ is not a minimizer.
- (d) Under the assumption, $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y}\|^2 - 2 \sum_g \mathbf{z}_g^\top \mathbf{b}_g + \sum_g \|\mathbf{b}_g\|^2$. So Eq. 30.3.1 separates into the problems $\min_{\mathbf{b}_g} \{\frac{1}{2} \|\mathbf{z}_g - \mathbf{b}_g\|^2 + \lambda w_g \|\mathbf{b}_g\|\}$, each strictly convex. Let $c = \lambda w_g$ and let $\mathbf{b}_g$ be given by Eq. 30.3.3. If $\|\mathbf{z}_g\| \leq c$, then $\mathbf{b}_g = \mathbf{0}$ and $\|\mathbf{z}_g - \mathbf{0}\| \leq c$. If $\|\mathbf{z}_g\| > c$, then $\mathbf{b}_g = (\|\mathbf{z}_g\| - c) \mathbf{z}_g / \|\mathbf{z}_g\| \neq \mathbf{0}$ and $\mathbf{z}_g - \mathbf{b}_g = c \mathbf{z}_g / \|\mathbf{z}_g\| = c \mathbf{b}_g / \|\mathbf{b}_g\|$. In both cases the one-group version of Eq. 30.3.2 holds.
- (e) As a function of $\mathbf{b}_g$, the criterion is $\frac{1}{2} \|\mathbf{r}_{(g)} - \mathbf{X}_g \mathbf{b}_g\|^2 + \lambda w_g \|\mathbf{b}_g\|$ plus a constant. With $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}$, this is the one-group problem of (d) with $\mathbf{z}_g$ replaced by $\mathbf{s}_g$.

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Block soft thresholding keeps the direction of the least squares vector $\mathbf{z}_g$ and reduces its length by λw_g : the lasso rule applied to a length. The weights come from (c): for pure noise and orthonormal groups, $\|\mathbf{X}_g^\top \mathbf{y}\|^2 / \sigma^2 \sim \chi^2(p_g)$ has mean p_g , so without $\sqrt{p_g}$ large groups would enter merely for being large (Exercise (A1)). Block coordinate descent with update (e) has every limit point a minimizer because the penalty is separable across blocks (Tseng 2001), so it converges when the minimizer is unique; without orthonormality the block problem has no closed form (Exercise (C1)).

30.3.2 Factors as groups

Orthonormalizing within groups also makes the group lasso independent of how a factor is coded.

Proposition 30.3.2. Invariance to the coding of a group

- (a) Let $\mathbf{U}_g$ be $p_g \times p_g$ orthogonal matrices, and replace every $\mathbf{X}_g$ by $\mathbf{X}_g \mathbf{U}_g$. Then $\hat{\boldsymbol{\beta}}$ minimizes the original criterion iff the vector with blocks $\mathbf{U}_g^\top \hat{\boldsymbol{\beta}}_g$ minimizes the new one. The fitted contributions $\mathbf{X}_g \hat{\boldsymbol{\beta}}_g$ are the same.
- (b) Suppose each group is orthonormalized within itself, so that $\mathbf{X}_g$ is an orthonormal basis of a subspace $\mathcal{V}_g$. Then the minimizers, expressed through the fitted contributions $\mathbf{X}_g \hat{\boldsymbol{\beta}}_g \in \mathcal{V}_g$, depend on the groups only through the subspaces $\mathcal{V}_g$. In particular, for a factor they do not depend on the choice of reference level or of contrasts.

Proof.

- (a) With $\mathbf{b}_g = \mathbf{U}_g \mathbf{c}_g$, we have $\mathbf{X}_g \mathbf{U}_g \mathbf{c}_g = \mathbf{X}_g \mathbf{b}_g$ and $\|\mathbf{c}_g\| = \|\mathbf{b}_g\|$. So the new criterion at $\mathbf{c}$ equals the old criterion at $\mathbf{b}$, and $\mathbf{b} \leftrightarrow \mathbf{c}$ is a bijection.
- (b) Two orthonormal bases $\mathbf{X}_g$ and $\mathbf{X}'_g$ of $\mathcal{V}_g$ satisfy $\mathbf{X}'_g = \mathbf{X}_g \mathbf{U}_g$ with $\mathbf{U}_g = \mathbf{X}_g^\top \mathbf{X}'_g$ orthogonal. Apply (a). For a factor with k levels, the centred indicator columns of any reference coding, and the centred columns of any other full-rank contrast coding, span the same $(k-1)$ -dimensional subspace: the vectors that are constant within levels and

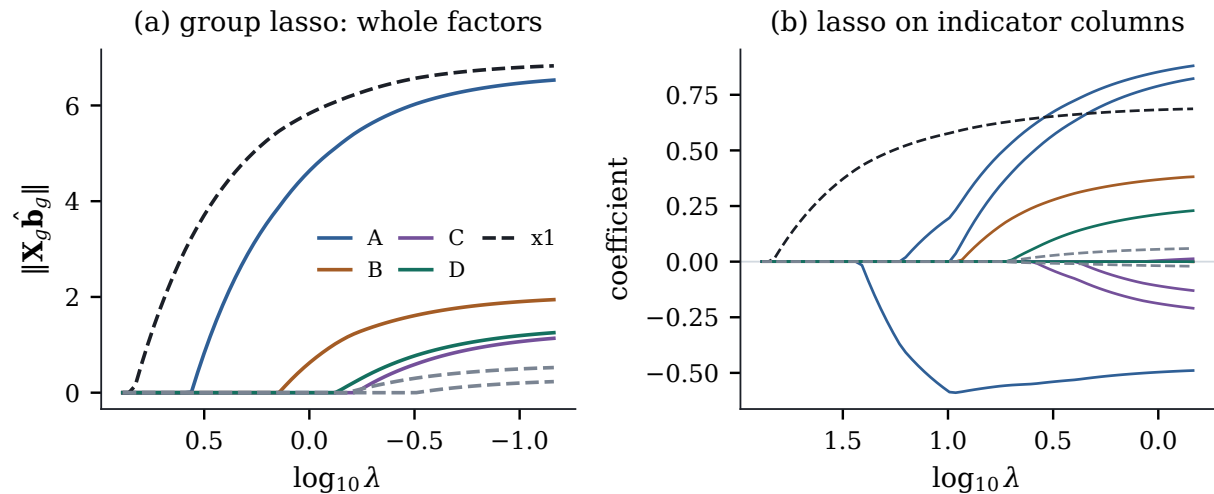


Figure 30.3.1: (a) Group lasso path for [Example \(Selecting factors\)](#): the length of each group's fitted contribution $\|\mathbf{X}_g \hat{\mathbf{b}}_g\|$ against $\log_{10} \lambda$. Factors enter as wholes. (b) The lasso path for the same data, with each indicator column penalized separately, coloured by factor. The dashed curves are the numeric regressors.

orthogonal to $\mathbf{1}$.

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With orthonormalized groups $\|\mathbf{b}_g\| = \|\mathbf{X}_g \mathbf{b}_g\|$: the penalty is the total length of the groups' fitted contributions, a coding-free quantity. Simon and Tibshirani (2012) argue for this version.

Example. Selecting factors

The script simulates $n = 120$ observations of four factors A, B, C, D (with 4, 3, 5 and 3 levels) and three numeric regressors; the response depends on A , weakly on B , and on x_1 . The group lasso path starts at $\lambda_{\max} = 6.876$, and groups enter in the order x_1, A, B, D . At $\lambda = 2.0$ only A and x_1 are selected, with $\|\mathbf{X}_A \hat{\mathbf{\beta}}_A\| = 2.834$ and $\|\mathbf{X}_{x_1} \hat{\mathbf{\beta}}_{x_1}\| = 4.861$. Another reference level gives the same fitted contributions to machine precision. The lasso on the raw indicator columns does not: at $\lambda = 6.0$ its fit moves by 8.6 per cent of its length when the reference levels change. Along its path the columns of A enter at values of λ spread over a factor of 2.8, and it admits its first column of a null factor (C or D) at a residual sum of squares of 112.0, a worse fit than the 110.7 that the group lasso reaches with both null factors still excluded ([Figure 30.3.1](#)).

```
import numpy as np

rng = np.random.default_rng(3003)
n = 120
levels = {"A": 4, "B": 3, "C": 5, "D": 3}
factors = {f: rng.integers(0, k, size=n) for f, k in levels.items()}
numeric = rng.normal(size=(n, 3))
effect_A = np.array([0.0, 1.0, -0.5, 0.8])
effect_B = np.array([0.0, 0.3, -0.3])
y = effect_A[factors["A"]] + effect_B[factors["B"]] + 0.6 * numeric[:, 0] + rng.
    normal(size=n)
y -= y.mean()
```

```

def indicator_columns(codes, k, reference=0):
    """Indicator columns for the levels other than the reference level."""
    return np.column_stack([(codes == l).astype(float) for l in range(k) if l !=
                             reference])

def orthonormal_group(Z):
    """Centre the columns, then an orthonormal basis of their span ( $Z = QR$ )."""
    Z = Z - Z.mean(axis=0)
    Q, _ = np.linalg.qr(Z)
    return Q

def build(reference=0):
    blocks = [orthonormal_group(indicator_columns(factors[f], k, reference))
               for f, k in levels.items()]
    blocks += [orthonormal_group(numeric[:, [j]]) for j in range(3)]
    return blocks

def group_lasso(blocks, y, lam, b=None, tol=1e-12, max_sweeps=100000):
    """ $(1/2) ||y - \sum_g X_g b_g||^2 + \lambda \sum_g \sqrt{p_g} ||b_g||$ , with  $X_g' X_g = I$ .
    """
    b = [np.zeros(Xg.shape[1]) for Xg in blocks] if b is None else [v.copy() for v
                                                                    in b]
    r = y - sum(Xg @ bg for Xg, bg in zip(blocks, b))
    for _ in range(max_sweeps):
        largest = 0.0
        for g, Xg in enumerate(blocks):
            s = Xg.T @ r + b[g] # block least squares on the
            # partial residual
            w = np.sqrt(Xg.shape[1])
            norm_s = np.linalg.norm(s)
            new = max(0.0, 1 - lam * w / norm_s) * s if norm_s > 0 else 0 * s
            r += Xg @ (b[g] - new)
            largest = max(largest, np.max(np.abs(new - b[g])))
            b[g] = new
        if largest < tol:
            return b
    raise RuntimeError("block coordinate descent did not converge")

blocks = build()
names = list(levels) + ["x1", "x2", "x3"]
lam = 2.0
b = group_lasso(blocks, y, lam)
for name, Xg, bg in zip(names, blocks, b):
    print(f"{name}: {Xg.shape[1]} columns, ||X_g b_g|| = {np.linalg.norm(Xg @ bg):.3f}")

```

[Run this code online](#)

The **sparse group lasso** adds an ℓ_1 term so that a selected group can have zero coefficients (Simon, Friedman, Hastie and Tibshirani, 2013).

30.3.3 Ordered coefficients: the fused lasso

When coefficients have a natural order (levels of an ordered factor, successive times), neighbours are expected to be similar. The **fused lasso** of Tibshirani, Saunders, Rosset, Zhu and Knight (2005) penalizes their differences:

$$Q(\mathbf{b}) = \frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda_1 \sum_{j=1}^p |b_j| + \lambda_2 \sum_{j=2}^p |b_j - b_{j-1}|. \quad (30.3.4)$$

It produces exact zeros and exact ties $\hat{\beta}_j = \hat{\beta}_{j-1}$, so the coefficient sequence is piecewise constant. The penalty is convex but not separable, so coordinate descent can stall at a tie. For an ordered factor, fusion merges adjacent levels the data cannot distinguish (Gertheiss and Tutz 2010). [Exercise \(B1\)](#) solves the smallest case.

30.3.4 Exercises

30.3.4.1 A. Check your understanding

Exercise. A1

Under orthonormal groups and $\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$, show that $P(\hat{\boldsymbol{\beta}}_g \neq \mathbf{0}) = P\{\chi^2(p_g) > \lambda^2 w_g^2 / \sigma^2\}$. With $w_g = 1$, compare these probabilities for $p_g = 1$ and $p_g = 10$ when $\lambda = 2\sigma$. What do the weights $\sqrt{p_g}$ achieve, and what do they not achieve?

Solution. By [Eq. 30.3.3](#), $\hat{\boldsymbol{\beta}}_g \neq \mathbf{0}$ iff $\|\mathbf{z}_g\| > \lambda w_g$. Here $\mathbf{z}_g = \mathbf{X}_g^\top \mathbf{y} \sim N_{p_g}(\mathbf{0}, \sigma^2 \mathbf{I})$, so $\|\mathbf{z}_g\|^2 / \sigma^2 \sim \chi^2(p_g)$. With $w_g = 1$ and $\lambda = 2\sigma$, the probabilities are $P\{\chi^2(1) > 4\} = 0.046$ and $P\{\chi^2(10) > 4\} = 0.947$, so a large null group almost always enters. With $w_g = \sqrt{p_g}$ the threshold is $4p_g$, four times the mean of $\chi^2(p_g)$. This removes the gross imbalance, but it does not equalize the probabilities: they are 0.046 and $P\{\chi^2(10) > 40\} = 1.7 \times 10^{-5}$. Equal null entry probabilities would need thresholds at a common quantile of $\chi^2(p_g)$.

30.3.4.2 B. Practice

Exercise. B1

Solve the fused lasso [Eq. 30.3.4](#) with $\mathbf{X} = \mathbf{I}_2$, $\lambda_1 = 0$ and $\lambda_2 = \lambda$. Show that $\hat{\beta}_1 = \hat{\beta}_2 = \frac{1}{2}(y_1 + y_2)$ if $|y_1 - y_2| \leq 2\lambda$. Otherwise, each estimate moves a distance λ from its observation towards the other.

Solution. The penalty is $\lambda|b_1 - b_2| = \lambda|\mathbf{a}^\top \mathbf{b}|$ with $\mathbf{a} = (1, -1)^\top$. Its subgradients are $\lambda s \mathbf{a}$ with $s \in \partial|\cdot|(b_1 - b_2)$. By Lemma 30.1.2, $y_1 - b_1 = \lambda s$ and $y_2 - b_2 = -\lambda s$. If $b_1 = b_2 = b$, adding the equations gives $b = \frac{1}{2}(y_1 + y_2)$, and then $s = (y_1 - y_2)/(2\lambda)$ must lie in $[-1, 1]$. If $|y_1 - y_2| > 2\lambda$, then $b_1 \neq b_2$, $s = \text{sign}(b_1 - b_2)$, and $b_1 = y_1 - \lambda s$, $b_2 = y_2 + \lambda s$. This is consistent when $s = \text{sign}(y_1 - y_2)$, since then $b_1 - b_2 = y_1 - y_2 - 2\lambda s$ has the sign of s .

Exercise. B2

Under an orthonormal design, compare the group lasso (one group of size 3, $w = \sqrt{3}$) with the lasso on the same three columns, for $\mathbf{z} = (2, 0.3, -0.2)^\top$ and $\lambda = 1$. Which coefficients

are zero in each? Show that, under an orthonormal design, the group lasso never sets a single coordinate of a selected group to zero unless the corresponding $z_j = 0$.

30.3.4.3 C. Going deeper

Exercise. C1

Drop the assumption $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}$. Show that a nonzero solution of the block problem $\min_{\mathbf{b}} \frac{1}{2} \|\mathbf{r} - \mathbf{X}_g \mathbf{b}\|^2 + c \|\mathbf{b}\|$ is $\mathbf{b} = (\mathbf{X}_g^\top \mathbf{X}_g + (c/t)\mathbf{I})^{-1} \mathbf{X}_g^\top \mathbf{r}$, where $t = \|\mathbf{b}\|$ is the root of a one-dimensional equation. Show that this equation has a unique root when $\|\mathbf{X}_g^\top \mathbf{r}\| > c$, so the block update is a ridge regression (Theorem 27.2.2) with a data-dependent parameter.